Autograd in NumPy

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1 Introduction

In this article, we will implement a simple autograd system in NumPy. Autograd is a technique for automatically computing gradients of functions, which is essential for training machine learning models without manually deriving gradients. We will define a class for tensors that can track operations and compute gradients, as well as some basic operations like addition and matrix multiplication. We will also implement a simple ReLU activation function and demonstrate how to use our autograd system to compute gradients for a simple neural network.

2 Autograd Implementation

The fundamental data structure in our autograd system will be the `Tensor` class. This class